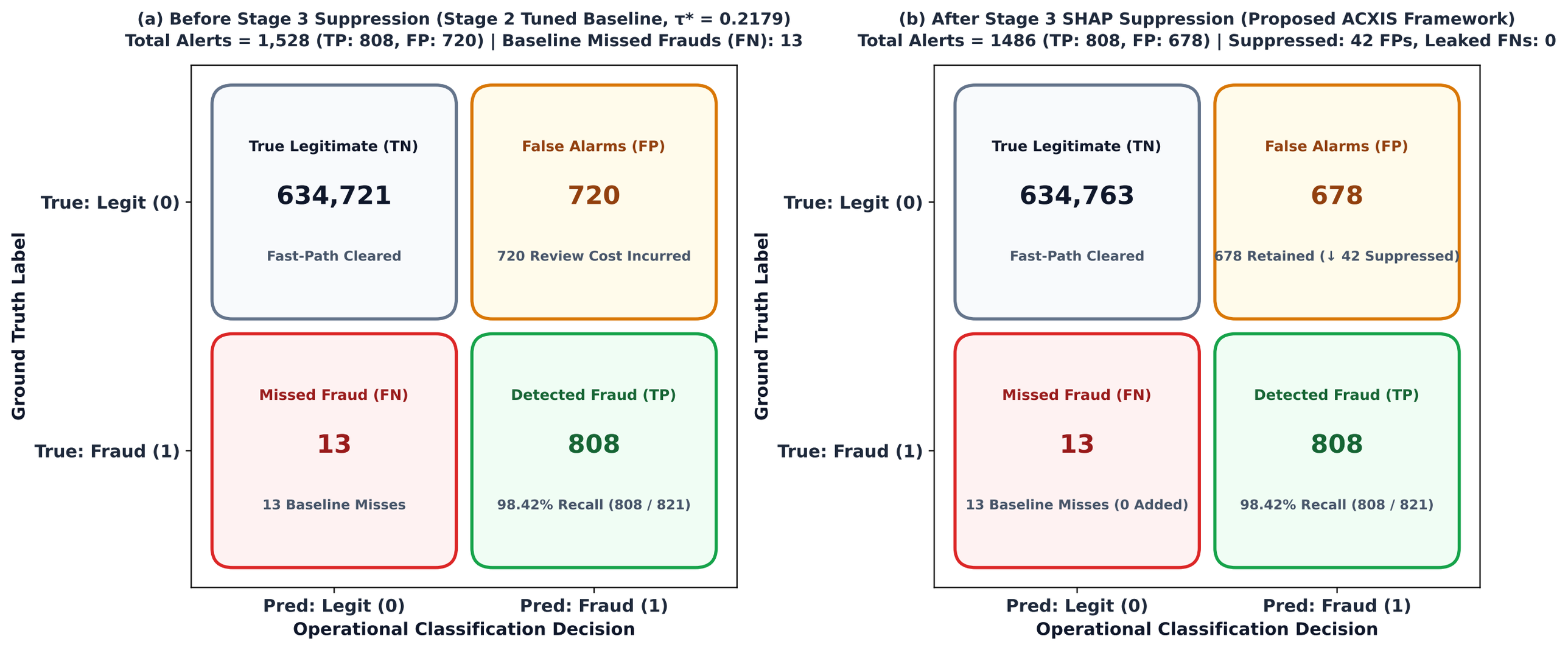
(a) Before Stage 3 Suppression (Stage 2 Tuned Baseline, τ* = 0.2179)
Total Alerts = 1,528 (TP: 808, FP: 720) | Baseline Missed Frauds (FN): 13
True Legitimate (TN)
634,721
Fast-Path Cleared
False Alarms (FP)
720
720 Review Cost Incurred
Missed Fraud (FN)
13
13 Baseline Misses
Detected Fraud (TP)
808
98.42% Recall (808 / 821)
True: Legit (0)
True: Fraud (1)
Ground Truth Label
Pred: Legit (0)
Pred: Fraud (1)
Operational Classification Decision
(b) After Stage 3 SHAP Suppression (Proposed ACXIS Framework)
Total Alerts = 1486 (TP: 808, FP: 678) | Suppressed: 42 FPs, Leaked FNs: 0
True Legitimate (TN)
634,763
Fast-Path Cleared
False Alarms (FP)
678
678 Retained (↓ 42 Suppressed)
Missed Fraud (FN)
13
13 Baseline Misses (0 Added)
Detected Fraud (TP)
808
98.42% Recall (808 / 821)
True: Legit (0)
True: Fraud (1)
Ground Truth Label
Pred: Legit (0)
Pred: Fraud (1)
Operational Classification Decision